09 The Diversity Room

Measuring a community

CORE PATH · BSC	EXTENSION · MSC	BEFORE THIS ROOM
about 36 min	**+ 30 min**	Nothing essential. The Keystone Room (08) shows the kind of change these numbers are used to report.
steps 1–4	steps 5–7	

OPEN THE ROOM

`https://evolinus.github.io/zooe/teaching/population-laboratories/` — then the Diversity tab

A real community has dozens of species, and ecologists sum it up in a number. This room builds a community, samples it the way a field ecologist must, and lets it drift with no species better than another. This guide asks what each number measures, how far a sample can be trusted, and what keeps a community diverse.

LEARNING OBJECTIVES

After the core path you should be able to

- Distinguish richness from evenness, and explain why both make up diversity.
- Compare richness, Shannon's and Simpson's effective numbers, and say how each weighs rare species.
- Explain why richness estimated from a sample is biased low, and why comparing samples of different size is unsafe.
- Show that neutral drift removes species and that immigration can stop the loss.

The extension adds

- Calculate H', D and the Hill numbers by hand and check them.
- Read a species-accumulation curve and relate it to sampling effort.
- Explain how community size J and immigration m set the diversity a neutral community settles at.

SUGGESTED ROUTES

- **40-minute practical (BSc):** steps 1–4 and questions 1–3.
- **Field-course preparation:** steps 3 and 6 before students collect their own samples.
- **MSc:** steps 5–7 and questions 4–6 (about 30 minutes).

How to work through it. Every step asks you to *predict* before you press anything, and to *explain* afterwards. Write the prediction down and commit to it: a wrong prediction you can account for teaches more than a right one you guessed. Runs are random, so where a step says so, run it more than once before you decide what happened.

1 Same species, different diversity

8 min

SET UP

Step 1 of the room, $S = 8$. Preset **Even**, then **One dominant**.

PREDICT FIRST

In the even community, what will richness, Shannon's and Simpson's effective numbers be? In the one-dominant community, which of the three will fall furthest?

RUN

Choose each preset and read the panels.

LOOK AT

Effective number of species: richness, ${}^1D = e^{H'}$ and ${}^2D = 1/D$.

EXPLAIN

Both communities hold the same eight species. Why are they so different in diversity?

2 Add a rare species

8 min

SET UP

Preset **One dominant**.

PREDICT FIRST

You add one rare species. Which of the three numbers will change most, and which least?

RUN

Press **Add a rare species**. Then drag its bar down to almost nothing.

EXPLAIN

Choosing an index is choosing how much the rare species should count. Which index would you use to report a conservation success for a rare species, and which to describe the community's common core?

3 Sampling the community

12 min

SET UP

Step 2 of the room. Preset **Geometric** (S = 8). Sample size n = 20.

PREDICT FIRST

In a sample of 20 individuals, how many of the 8 species will you find? Over 100 such samples, which estimate (richness, Shannon, Simpson) will be furthest below the truth on average?

RUN

Press **Take a sample**, then **Take 100 samples**.

LOOK AT

Species found as you count and *A hundred samples against the truth*.

EXPLAIN

Why can a sample miss a species but never invent one? What does that mean for richness comparisons between surveys of different effort?

4 Drift, with and without immigrants

8 min

SET UP

Step 3 of the room: J = 500, 400 generations. Immigration m = **0**, then m = **0.01**.

PREDICT FIRST

No species is better than any other. Will the community keep all its species? What will a trickle of immigration change?

RUN

Run each.

LOOK AT

Who holds the community and *Diversity over time*.

EXPLAIN

Why does diversity drain away with no cause? Which room in the Evolutionary Laboratories does this mirror?

5 By hand

10 min

SET UP

Drag the bars to build a four-species community with abundances 40, 30, 20, 10.

PREDICT FIRST

Calculate the shares p_i, Shannon's $H' = -\Sigma p_i \ln p_i$, Simpson's $D = \Sigma p_i^2$, and the two effective numbers $e^{H'}$ and $1/D$, and Pielou's J'.

RUN

Read the room's values.

EXPLAIN

Why are effective numbers easier to compare than the raw indices?

6 Effort and accumulation

10 min

SET UP

Geometric preset. Sample size n = 20, then 100, then 400.

PREDICT FIRST

How will the average richness estimate change with n? Will Simpson's estimate change much?

RUN

Take 100 samples at each n.

EXPLAIN

Use the accumulation curve to explain why richness is so sensitive to effort. Name an estimator that tries to correct for unseen species.

7 Size and connection

10 min

SET UP

Step 3 with m = 0.005. J = 200, then J = 2000.

PREDICT FIRST

Which community will hold more species at the end?

RUN

Run each.

EXPLAIN

The room says diversity depends on how large and how connected a place is, as well as on who lives there. Explain why, using the balance between drift and immigration.

DISCUSSION AND WORKSHEET QUESTIONS

1. BSc Two meadows each have 20 plant species. In one, a single grass makes up 90% of the plants. Which is more diverse, and by which measure?
2. BSc A student reports that site A has more species than site B. Site A was sampled for twice as long. What should you ask before accepting the conclusion?
3. BSc Why might a manager care about Shannon's number rather than raw richness?
4. MSc Hubbell's neutral theory assumes all species are equivalent. What evidence would you look for to test it?
5. MSc Explain why Simpson's Σp_i^2 is the probability that two individuals belong to the same species, and connect it to heterozygosity.
6. MSc Rarefaction is used to compare samples of different size. Explain its logic in terms of the accumulation curve.

CHECKING YOUR REASONING

Read this after you have written your own answers. These are what the model should show on most runs — not the only acceptable explanations.

- **Step 1.** Even: all three equal 8. One dominant: richness 8, Shannon's about 2.9, Simpson's about 1.8 — Simpson's falls furthest because it is set by the commonest species.
- **Step 2.** Richness rises by one; Shannon's by a little (about 0.8 when the new species has 20 individuals, less when it is dragged down); Simpson's least of all. Richness suits rare-species reporting; Simpson's describes the common core.
- **Step 3.** A sample of 20 usually finds 4–6 of the 8. Richness is the most biased low, Shannon's less so, Simpson's is nearly right.
- **Step 4.** With $m = 0$ species are lost one after another and eventually one takes over. A little immigration halts the slide. It is the Drift Room with species in place of alleles.
- **Step 5.** $p = 0.4, 0.3, 0.2, 0.1$; $H' \approx 1.28$, $e^{H'} \approx 3.6$; $D = 0.30$, $1/D \approx 3.3$; $J' \approx 0.92$.
- **Step 6.** Richness climbs towards 8 as n grows; Simpson's barely moves. Chao1 is one estimator of unseen species.
- **Step 7.** $J = 2000$ holds more species: drift is slower in a larger community, so immigration keeps up with the losses.